



Nova School of Business & Economics

**The Causal Effect of Minimum Tick Size Increases
on Trading Volume (Measured by Turnover)**

Group Assignment Research Methods for Finance

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List of Abbreviations

CRSP	Center for Research in Security Prices
DiD	Difference-in-Differences
G1	Pilot Group 1
G2	Pilot Group 2
G3	Pilot Group
H1	Hypothesis 1 (Average Treatment Effect)
H2	Hypothesis 2 (Heterogeneous Treatment Effect)
ITT	Intention-to-Treat
L_abs_ret	One-Day Lagged Absolute Return
Ln	Natural Logarithm
PERMNO	Permanent Identification Number (CRSP Stock Identifier)
SEC	Securities and Exchange Commission
vol	Daily Share Volume
shROUT	Shares Outstanding

1 Hypotheses Development

This section develops the theoretical framework guiding the empirical analysis of the SEC Tick Size Pilot Program. Building on market microstructure theory, testable predictions are formulated regarding the effect of an exogenous tick size increase on trading volume. In the empirical analysis, trading activity is measured by daily share turnover, defined as share volume scaled by shares outstanding. The analysis considers both the average treatment effect and potential heterogeneity arising from pre-treatment spread differences.

1.1 Research Question

This paper examines the causal effect of a minimum tick size increase on equity trading volume. The SEC Tick Size Pilot Program, implemented over the course of October 2016, raised the minimum quoting increment from \$0.01 to \$0.05 for approximately 1,200 randomly selected small-capitalization stocks, while a control group continued trading under the existing \$0.01 regime (Albuquerque et al., 2020). Treated stocks were assigned to three pilot groups. Group 1 stocks were required to quote in \$0.05 increments but could continue to execute trades at finer increments. Group 2 stocks faced both the \$0.05 quoting and trading requirement, eliminating sub-increment execution. Group 3 stocks were subject to all requirements imposed on Group 2 and additionally to a trade-at prohibition, which prevents dark venues from executing orders at the protected displayed quote unless they offer a price improvement of at least \$0.05, raising the cost of off-exchange trading. The stratified random assignment provides a natural experiment supporting causal identification within a difference-in-differences framework. Two central research questions are examined. First, does the tick size increase reduce trading volume across treatment groups relative to the control group, and do the effects differ across groups with escalating regulatory requirements? Second, is the decline in trading activity concentrated among stocks whose pre-treatment spreads were narrower than \$0.05, meaning that the new tick size forces their spreads to widen?

1.1 Economic Framework

The following premises underpin our hypotheses.

P1. Trading volume reflects investors' willingness to transact and is negatively related to transaction costs. Higher bid-ask spreads reduce expected net gains from trade and eliminate marginal transactions, lowering turnover (Amihud & Mendelson, 1986; Chordia et al., 2001).

P2. The minimum tick size imposes a regulatory lower bound on quoted spreads. Because prices cannot be improved by less than one tick, a wider tick restricts price competition in the limit order book and prevents spreads from falling below the floor (Harris, 1994). Evidence from decimalization shows that reducing the tick compresses spreads, confirming that it can bind in equilibrium (Bessembinder, 2003).

P3. Investors trade only when expected gains exceed round-trip transaction costs. An increase in the spread floor therefore renders previously profitable trades unviable, reducing observed trading volume.

P4. The \$0.05 minimum tick binds only for stocks whose pre-treatment spreads were below \$0.05. For these stocks, the regulation forces spreads above prior equilibrium levels; for stocks already quoting above \$0.05, the constraint is non-binding and spreads remain unchanged (Albuquerque et al., 2020).

The following assumptions are required for causal interpretation.

A1 (Exogeneity). Treatment assignment was determined by the SEC's stratified random sampling procedure and is therefore exogenous to contemporaneous firm fundamentals. Balance tests in Albuquerque et al. (2020) confirm comparability between treated and control stocks prior to implementation. Firm fixed effects absorb any remaining time-invariant heterogeneity.

A2 (Parallel Trends). In the absence of the pilot, treated and control stocks would have followed parallel trends in trading volume. The difference-in-differences estimator attributes post-pilot divergence to the treatment; we assess this assumption through pre-trend tests.

A3 (Predetermined Bindingness). Pre-treatment average quoted spreads provide a valid proxy for whether the \$0.05 floor is binding and are free of anticipation effects. Any measurement error would attenuate heterogeneous treatment estimates toward zero, making significant effects conservative.

1.2 Hypothesis 1: Average Treatment Effect

If the friction channel dominates the liquidity-supply channel, treated stocks should experience lower trading volume relative to control stocks in the post-pilot period, despite the intended benefits of greater market maker participation.

H1: The regulatory increase in the minimum tick size from \$0.01 to \$0.05 reduces trading volume for treated firms relative to control firms.

The corresponding null hypothesis is that the tick size increase does not affect trading volume.

1.3 Hypothesis 2: Heterogeneous Effects by Pre-Pilot Spread Narrowness

The reduction in volume should not be uniform across treated stocks. Stocks that traded with narrow spreads before the pilot, that is, well below \$0.05, are most exposed to the regulatory change because the new minimum tick forces a larger increase in transaction costs. Stocks already trading with spreads at or above \$0.05 are largely unaffected, since the new tick does not require spreads to widen.

H2: The negative effect of the tick size increase on trading volume is larger for stocks whose pre-pilot quoted spreads were below \$0.05.

The corresponding null hypothesis is that the effect of the tick size increase on volume does not vary with pre-pilot spread narrowness.

2 Empirical Methodology

This section presents the empirical strategy used to estimate the causal impact of the SEC Tick Size Pilot Program on trading activity. The analysis employs a difference-in-differences framework with firm and time fixed effects to identify both the average treatment effect and heterogeneous effects based on pre-treatment spread constraints.

2.1 Identification Strategy

The causal effect of the minimum tick size increase on trading volume is identified using a difference-in-differences framework, exploiting the exogenous variation generated by the SEC's stratified random assignment of eligible small-capitalization stocks into three treatment groups and a control group (Albuquerque et al., 2020). Because assignment was determined by regulatory procedure rather than firm choice, treatment status is orthogonal to contemporaneous firm fundamentals. The estimator compares changes in trading activity across three periods: a pre-treatment baseline (January 1, 2016 through October 16, 2016 for Groups 1 and 2, and January 1, 2016 through October 30, 2016 for Group 3), the active pilot (October 17, 2016 through September 28, 2018 for Groups 1 and 2, and October 31, 2016 through September 28, 2018 for Group 3), and a post-pilot window (September 29, 2018 through April 30, 2019) during which tick sizes reverted to \$0.01. The validity of this approach rests on the parallel trends assumption, which is supported by the random assignment of stocks to groups. Separate treatment coefficients are estimated for each pilot group to test whether more binding regulatory constraints generate correspondingly larger reductions in trading activity. The post-pilot period is included to assess whether volume effects persist after the regulatory change is reversed, which would indicate lasting structural shifts in trading activity rather than a purely mechanical response to the tick size constraint.

2.2 Baseline Specification

The baseline specification for Hypothesis 1 allows for a separate treatment effect in each pilot group across both the active pilot period and the post-pilot reversal period:

$$\begin{aligned} \text{Volume}_{it} = & \alpha + \beta_1(G_{1i} \times \text{TreatmentPeriod}_t) + \beta_2(G_{2i} \times \text{TreatmentPeriod}_t) \\ & + \beta_3(G_{3i} \times \text{TreatmentPeriod}_t) + \lambda_1(G_{1i} \times \text{PostPilot}_t) + \lambda_2(G_{2i} \times \text{PostPilot}_t) \\ & + \lambda_3(G_{3i} \times \text{PostPilot}_t) + \theta' X_{i,t} + \gamma_i + \delta_t + \varepsilon_{it} \end{aligned}$$

where Volume_{it} is the natural logarithm of daily share turnover for stock i on day t ; G_{1i} , G_{2i} , and G_{3i} are pilot group indicators; TreatmentPeriod_t equals one during the active pilot window with the pre-treatment period as the omitted baseline; PostPilot_t equals one after September 28, 2018; γ_i and δ_t denote firm and time fixed effects respectively; and $X_{i,t}$ is a vector of time-varying controls defined below. Standard errors are clustered at the firm level to account for serial correlation in residuals within

stocks over time. The coefficients β_1, β_2 , and β_3 capture the average volume change for each pilot group relative to control during the active pilot. Under H1, we predict $\beta_1, \beta_2, \beta_3 < 0$ and $|\beta_3| \geq \max(|\beta_1|, |\beta_2|)$, reflecting a dose-response relationship. The post-pilot coefficients λ_1, λ_2 and λ_3 test for persistence of the volume effect; under full reversion, these should be statistically indistinguishable from zero.

2.3 Extended Specification

To test Hypothesis 2, the baseline specification is augmented with triple interaction terms for both the active pilot period and the post-pilot period:

$$\begin{aligned} \text{Volume}_{it} = & \alpha + \sum_{k=1}^3 \beta_k (G_{ki} \times \text{TreatmentPeriod}_t) + \sum_{k=1}^3 \lambda_k (G_{ki} \times \text{PostPilot}_t) \\ & + \sum_{k=1}^3 \gamma_k (G_{ki} \times \text{TreatmentPeriod}_t \times \text{SmallSpread}_i) \\ & + \sum_{k=1}^3 \phi_k (G_{ki} \times \text{PostPilot}_t \times \text{SmallSpread}_i) + \theta' X_{it} + \gamma_i + \delta_t + \varepsilon_{it} \end{aligned}$$

Here, SmallSpread_i is a binary indicator equal to one for stocks whose pre-treatment average dollar quoted spread is strictly below \$0.05. The coefficients β_k capture the volume change for unconstrained stocks in each pilot group relative to control during the active pilot, while λ_k captures whether that effect persists in the post-pilot period. The coefficients γ_k capture the additional differential effect for constrained stocks during the active pilot over and above β_k , and ϕ_k analogously captures any differential persistence in the post-pilot period. The total treatment effect during the active pilot for a constrained stock in group k is therefore $\beta_k + \gamma_k$, while for an unconstrained stock it is simply β_k . Under H2, the predicted signs are $\gamma_1 < 0$, $\gamma_2 < 0$, and $\gamma_3 < 0$. Stocks with pre-treatment spreads at or above \$0.05 serve as an implicit placebo subgroup within each treated group, for which no additional differential volume effect is expected.

2.4 Control Variables

The inverse of the midpoint price (Inv_Price) is included as a time-varying control, capturing the mechanical relationship between price level and the relative size of the tick increment. Contemporaneous return volatility is excluded because the pilot directly widens spreads and thus affects short-term volatility, making it an endogenous outcome of the treatment, such that including it would introduce bad-control bias. Instead, we include the one-day lagged absolute return (L_abs_ret), which is predetermined relative to contemporaneous volume and captures intertemporal volatility persistence without introducing endogeneity. Firm and time fixed effects absorb time-invariant cross-sectional heterogeneity and aggregate market-wide shocks respectively.

3. Data

This section describes the data sources, sample construction, and variable definitions underlying the empirical analysis. It outlines the filtering procedures implemented to ensure a clean identification of the regulatory shock and explains the construction of the key treatment and liquidity measures. The resulting dataset forms the foundation for the difference-in-differences estimations presented below.

3.1 Data Construction and Sample Selection

We use daily stock-level data from the Center for Research in Security Prices (CRSP) over the period January 1, 2016 to April 30, 2019, consistent with Albuquerque et al. (2020). Stocks are identified using the CRSP PERMNO and merged with the SEC Tick Size Pilot treatment-control assignment list. The final sample consists exclusively of firms included in the SEC experiment.

To ensure that the analysis isolates the causal impact of the tick size increase rather than confounding corporate events, several filters are applied. Firms that undergo mergers or acquisitions during the sample period are fully excluded, as such events generate abnormal trading activity that could bias estimated treatment effects. Observations associated with delistings (delisting codes between 400 and 599) are removed to mitigate distortions arising from terminal trading behavior. Additionally, only ordinary common shares with share codes of 10 or 11 are retained.

Regarding penny stocks, we deliberately depart from applying a strict firm-level exclusion for stocks that ever drop below an absolute price of \$1 during the pilot. Instead, we exclude only the specific daily observations where the absolute price falls below \$1. Removing an entire firm from the panel simply because its price temporarily breaches this threshold introduces severe selection bias. Because the tick size increase inherently exerts downward pressure on stock prices, dropping entire firms would systematically eliminate the very stocks most adversely affected by the treatment. This violates the Intention-To-Treat (ITT) principle and compromises the exogeneity of the SEC's randomized assignment. Empirically, this selection bias artificially diminishes the statistical power of our second hypothesis: when enforcing the firm-level drop, the triple-interaction coefficients for constrained stocks lose conventional significance. Conversely, by employing the daily-level ITT approach, we effectively filter out extreme microstructure noise while preserving the sample's vital variance, yielding strong and statistically significant heterogeneous effects (Group 1: -0.169, $p = 0.031$; Group 3: -0.187, $p = 0.054$).

Continuous trading variables are further cleaned by removing observations with non-positive volume or shares outstanding, as well as cases in which the quoted ask does not exceed the bid. To limit the influence of extreme outliers, turnover, volume, spreads, absolute returns and inversed prices are winsorized at the 1st and 99th percentiles. Finally, to construct the SmallSpread indicator used in the heterogeneous analysis, each stock's average dollar quoted spread is computed over the six-month pre-implementation window prior to its activation date. Stocks with an average spread strictly below \$0.05

are classified as constrained, consistent with the binding-threshold definition in Albuquerque et al. (2020).

3.2 Variable Construction

The primary dependent variable is daily trading activity measured at the stock–day level. Consistent with the microstructure literature, trading activity is captured by share turnover, defined as the ratio of daily share volume (vol) to shares outstanding (shout). Turnover provides a scale-adjusted measure of trading intensity that facilitates comparability across firms of different sizes. The main outcome variable, $\ln(\text{Turnover}_{it})$, is constructed as the natural logarithm of daily turnover after winsorization. Logarithmic transformation reduces skewness and allows estimated coefficients to be interpreted approximately as percentage changes. For descriptive purposes, the logarithm of share volume is also computed, though it is not used in the baseline regressions.

Quoted spreads are constructed as the daily difference between the best ask and best bid prices, $\text{Spread}_{it} = \text{Ask}_{it} - \text{Bid}_{it}$. Observations in which the quoted ask does not exceed the bid are removed to ensure economically meaningful spread values. The spread measure serves two purposes: it underlies the construction of the binding-constraint indicator and provides descriptive evidence on pre-treatment liquidity conditions.

The key treatment variables correspond to the SEC pilot group assignments. Indicator variables $G1_i$, $G2_i$, and $G3_i$ identify stocks assigned to each of the three treatment groups, while control stocks constitute the omitted category. The treatment period indicator, TreatmentPeriod_t , equals one during the pilot implementation window (October 17, 2016 to September 28, 2018 for Groups 1 and 2; October 31, 2016 to September 28, 2018 for Group 3) and zero otherwise. These indicators generate the difference-in-differences interactions that identify the average treatment effects.

To capture heterogeneity in exposure to the regulatory change, a pre-treatment binding constraint indicator, SmallSpread_i , is constructed. For each stock, the average dollar quoted spread is computed over the six-month pre-implementation window corresponding to its activation date. The firm-level pre-period average is then assigned to all observations for that stock. Stocks whose average pre-treatment spread is strictly below \$0.05 are classified as constrained. This classification reflects whether the new \$0.05 minimum tick is likely to bind relative to the stock’s prior equilibrium spread.

Two time-varying control variables are included in all specifications. The first is the inverse of the daily midpoint price, defined as $\text{Inv_Price}_{it} = \frac{1}{(\text{Ask}_{it} + \text{Bid}_{it})/2}$, which captures the mechanical relationship between the price level and the proportional size of the minimum tick increment, as lower-priced stocks face a larger relative spread floor. Including this control ensures that estimated treatment effects are not driven by contemporaneous price-level variation. The second control variable is the one-day lagged absolute return (L_abs_ret), which is predetermined relative to contemporaneous volume and controls

for short-term intertemporal volatility persistence, as turbulent trading days tend to be followed by elevated trading activity, without introducing endogeneity concerns. Contemporaneous return volatility is deliberately excluded from the specification. Because the pilot directly widens bid-ask spreads, contemporaneous volatility constitutes an endogenous outcome of the regulatory change, and its inclusion would introduce bad-control bias by absorbing part of the treatment effect. Firm and date fixed effects absorb time-invariant cross-sectional heterogeneity and aggregate market-wide shocks respectively, rendering no further controls necessary.

3.3 Summary Statistics

Table 1 reports summary statistics for the control group and the three pilot treatment groups during a 6-month pre-implementation period, from March 1 to August 31, 2016 (Albuquerque et al., 2020). Restricting the sample to this pre-treatment window ensures that the statistics reflect baseline characteristics prior to the regulatory intervention and are not mechanically influenced by the tick size increase.

Across all four groups, trading activity is economically comparable before the pilot's implementation. The mean log turnover is 1.113 in Group 1, 1.064 in Group 2, and 1.113 in Group 3, with the control group exhibiting a mean of 1.104. Similarly, average log trading volume varies modestly across groups, remaining tightly clustered around 11.02-11.13. These similarities suggest no systematic pre-treatment differences in baseline trading intensity between treated and control stocks.

Quoted spread measures display the same pattern of comparability. The average daily spread equals 8.3 cents in the control group, compared to 7.8 cents in Group 1, 7.9 cents in Group 2, and 8.2 cents in Group 3. Firm-level pre-treatment spreads are likewise closely aligned across groups, with differences around half of a cent. The narrow dispersion in both contemporaneous and historical spread measures is consistent with the SEC's stratified random assignment procedure and supports the assumption that treatment status is exogenous to prior liquidity conditions.

The proportion of stocks classified as constrained (SmallSpread) is also remarkably similar across groups. Approximately three-quarters of stocks in each group exhibit pre-treatment average spreads below the \$0.05 threshold, with shares ranging from 73.9% to 74.7%. This balance is particularly important for the heterogeneous analysis, as it indicates that the distribution of binding constraints is not systematically correlated with treatment assignment.

Overall, the pre-treatment statistics reveal no economically meaningful baseline imbalances across control and treatment groups. The similarity in trading activity, spreads, and constraint classification prior to implementation strengthens the credibility of the difference-in-differences design and supports the parallel trends assumption underlying the empirical strategy. turnover, volume, spreads, absolute returns and inversed prices.

Table 1: Pre-Treatment Summary Statistics (March 1 to August 31, 2016)

Panel A: Control Group

Variable	Count	Mean	Variance	Std. Dev.	Min	Max	Sum
ln_turnover	108,011	1.104	1.897	1.377	-3.602	3.855	119,288
ln_volume	108,011	11.092	3.828	1.957	4.828	14.539	1,198,082
Inv_Price	108,011	.097	.010	.097	.006	.669	10,442
abs_ret	108,011	.018	0	.019	0	.104	1,918
spread	108,011	.083	.047	.218	.010	1.720	8,925
firm_pre_spread	108,011	.081	.037	.191	.010	1.705	8,715
SmallSpread	108,011	.746	.190	.435	0	1	80,581

Panel B: Pilot Group 1

Variable	Count	Mean	Variance	Std. Dev.	Min	Max	Sum
ln_turnover	37,324	1.113	1.763	1.328	-3.602	3.855	41,549
ln_volume	37,324	11.129	3.777	1.943	4.828	14.539	415,392
Inv_Price	37,324	.098	.010	.099	.006	.669	3,658
abs_ret	37,324	.017	0	.019	0	.104	643
spread	37,324	.078	.043	.208	.010	1.720	2,894
firm_pre_spread	37,324	.076	.034	.185	.010	1.702	2,820
SmallSpread	37,324	.739	.193	.439	0	1	27,597

Panel C: Pilot Group 2

Variable	Count	Mean	Variance	Std. Dev.	Min	Max	Sum
ln_turnover	36,484	1.064	2.069	1.439	-3.602	3.855	38,810
ln_volume	36,484	11.021	4.175	2.043	4.828	14.539	402,105
Inv_Price	36,484	.107	.013	.114	.006	.669	3,919
abs_ret	36,484	.018	0	.019	0	.104	656
spread	36,484	.079	.038	.194	.010	1.720	2,868
firm_pre_spread	36,484	.079	.029	.171	.010	1.613	2,897
SmallSpread	36,484	.746	.190	.435	0	1	27,211

Panel D: Pilot Group 3

Variable	Count	Mean	Variance	Std. Dev.	Min	Max	Sum
ln_turnover	34,337	1.113	1.812	1.346	-3.602	3.855	38,222
ln_volume	34,337	11.118	3.592	1.895	4.828	14.539	381,746
Inv_Price	34,337	.097	.010	.101	.006	.669	3,343
abs_ret	34,337	.017	0	.018	0	.104	584
spread	34,337	.082	.052	.228	.010	1.720	2,811
firm_pre_spread	34,337	.079	.040	.200	.010	1.657	2,706
SmallSpread	34,337	.747	.189	.435	0	1	25,656

Summary statistics for the period March 1 to August 31, 2016 are reported following Albuquerque et al. (2020). Turnover, volume, spreads, absolute returns, and inverse prices are winsorized at the 1st and 99th percentiles.

4 Results

This section presents and discusses the results of the study. Table 2 reports the regression results for both hypotheses, with $\ln_turnover$ as the dependent variable.

Table 2: Regression Results for H1 and H2 (Dependent Variable: $\ln(\text{Turnover})$)

Panel A: Regression Results for H1 (Dependent Variable: $\ln(\text{Turnover})$)

Variable	Coefficient	Std. Err.	t	P> t	[95% Conf. Interval]	
G1 × Treatment Period	−0.088436	0.0328133	−2.70	0.007	−0.1527948	−0.0240772
G2 × Treatment Period	−0.0662199	0.0358505	−1.85	0.065	−0.1365357	0.004096
G3 × Treatment Period	−0.0954259	0.0373679	−2.55	0.011	−0.1687178	−0.022134
G1 × Post-Pilot	−0.0585513	0.0443249	−1.32	0.187	−0.1454884	0.0283859
G2 × Post-Pilot	−0.0995335	0.0464536	−2.14	0.032	−0.1906459	−0.0084211
G3 × Post-Pilot	0.0031071	0.0470065	0.07	0.947	−0.0890896	0.0953037
Inverse Price	−1.060827	0.1555809	−6.82	0.000	−1.365977	−0.7556768
Lagged Absolute Return	7.811796	0.1163718	67.13	0.000	7.583549	8.040043
Constant	1.21856	0.0174179	69.96	0.000	1.184397	1.252723

Panel B: Regression Results for H2 (Dependent Variable: $\ln(\text{Turnover})$)

(Std. err. adjusted for 1,700 clusters in permno)						
Variable	Coefficient	Robust std. err.	t	P> t	[95% conf. interval]	
					Lower	Upper
G1 × Treatment Period	0.0370424	0.0762318	0.49	0.627	−0.1124756	0.1865604
G2 × Treatment Period	0.0548058	0.0867653	0.63	0.528	−0.1153723	0.2249839
G3 × Treatment Period	0.0447011	0.0935399	0.48	0.633	−0.1387644	0.2281666
G1 × Post-Pilot	0.0240655	0.1040801	0.23	0.817	−0.1800731	0.2282042
G2 × Post-Pilot	0.0521394	0.1077217	0.48	0.628	−0.1591417	0.2634204
G3 × Post-Pilot	0.2269879	0.1180332	1.92	0.055	−0.0045178	0.4584936
G1 × Treatment × Small Spread	−0.1690721	0.0784224	−2.16	0.031	−0.3228867	−0.0152575
G2 × Treatment × Small Spread	−0.1607172	0.0899705	−1.79	0.074	−0.3371818	0.0157473
G3 × Treatment × Small Spread	−0.186671	0.0967912	−1.93	0.054	−0.3765136	0.0031716
G1 × Post-Pilot × Small Spread	−0.1110435	0.1080298	−1.03	0.304	−0.3229289	0.1008419
G2 × Post-Pilot × Small Spread	−0.2016958	0.1127759	−1.79	0.074	−0.4228901	0.0194985
G3 × Post-Pilot × Small Spread	−0.2990244	0.121643	−2.46	0.014	−0.5376103	−0.0604385
Inverse Price	−1.060384	0.1543352	−6.87	0.000	−1.363091	−0.7576772
Lagged Absolute Return	7.798215	0.1162055	67.11	0.000	7.570294	8.026136
Constant	1.218489	0.0172693	70.56	0.000	1.184618	1.25236

The dependent variable is $\ln(\text{Turnover})$ in both Panels A and B. G1, G2, and G3 indicate assignment to the SEC Tick Size Pilot treatment groups. Treatment equals one during the pilot implementation period. SmallSpread equals one if the firm's pre-treatment average quoted spread is below \$0.05. All regressions include firm and date fixed effects. Standard errors are clustered at the firm level and reported in parentheses. The table reports t-statistics, p-values, and 95% confidence intervals.

Table 2 reports the baseline difference-in-differences estimates of the impact of the minimum tick size increase on daily trading turnover. The dependent variable is the natural logarithm of daily share turnover. The interaction terms between the pilot group indicators (G1, G2, G3) and the Treatment Period indicator capture the change in turnover for treated stocks relative to control stocks during the pilot implementation window, while the Post-Pilot interactions measure whether effects persist after the program ended. In the heterogeneous specification, the triple interaction terms with SmallSpread identify whether the treatment effect differs for stocks whose pre-treatment quoted spreads were below

\$0.05. All specifications include firm and date fixed effects. Standard errors are clustered at the firm level to account for serial correlation within stocks over time.

4.1 Average Treatment Effect

Table 2 (Panel A) presents the baseline specification allowing for separate treatment effects for each pilot group. The estimated coefficients on the interaction terms between group indicators and the treatment period capture the average change in trading activity for treated stocks relative to the control group following the implementation of the pilot.

The estimated coefficient for Group 1 is -0.0884 and statistically significant at the 1 percent level ($p=0.007$). Because the dependent variable is log-transformed, the exact percentage effect is calculated as $e^{\beta} - 1$. This estimate implies an 8.46 percent decline in turnover for Group 1 stocks during the treatment period relative to the control group. For Group 2, the coefficient equals -0.0662 and is statistically significant at the 10 percent level ($p=0.065$), corresponding to a 6.41 percent reduction in trading activity. For Group 3, the coefficient equals -0.0954 and is statistically significant at the 5 percent level ($p=0.011$), implying a substantial 9.10 percent decline in turnover.

Furthermore, the post-pilot period indicators reveal that Group 1 and Group 3 fully reverted to control levels once the experiment concluded, while Group 2 shows a persistent and statistically significant decline of -0.0995 , suggesting the volume suppression did not fully reverse for this group.

The results are highly consistent with Hypothesis 1. Across all three pilot groups, the increase in the minimum tick size is associated with a statistically and economically significant reduction in trading volume. Exact declines between 6.4 and 9.1 percent in daily turnover indicate a notable contraction in trading activity for small capitalization stocks.

The ordering of the coefficients is also informative. Group 3, which faced the most restrictive trading conditions due to the trade-at prohibition, exhibits the largest reduction in turnover. Formal pairwise tests show that the differences between Group 3 and the other groups are not statistically significant at conventional levels (see Table 3 in the Appendix). Nevertheless, the pattern of point estimates aligns with the increasing intensity of regulatory constraints across groups. This ordering is consistent with the idea that progressively tighter trading restrictions reduce incentives to trade. Overall, the baseline evidence supports the friction channel described in the theoretical framework. By imposing a higher minimum quoting increment, the pilot increased effective transaction costs and reduced investors' willingness to trade.

4.2 Heterogeneous Effects by Pre-Treatment Spread

Table 2 (Panel B) augments the baseline specification with triple interaction terms between the treatment indicators and the pre-treatment binding constraint variable `SmallSpread`. This extension allows the treatment effect to differ depending on whether a stock's average pre-treatment quoted spread was below \$0.05. Under Hypothesis 2, the negative volume effect should be concentrated exclusively among stocks for which the new tick is likely to bind.

Once heterogeneous effects are introduced, the baseline treatment coefficients for Groups 1 to 3 become completely statistically insignificant during the active pilot period (p-values of 0.627, 0.528, and 0.633 for Groups 1, 2, and 3, respectively). These coefficients capture the effect for “placebo” stocks whose pre-treatment spreads were already at or above \$0.05. The absence of a significant effect for this subgroup confirms that the regulatory change did not materially alter trading activity for stocks that were unconstrained by the new tick size floor. During the post-pilot period, unconstrained stocks similarly show no meaningful effect for Groups 1 and 2 ($p=0.817$ and $p=0.628$). The only borderline exception is Group 3, where unconstrained stocks exhibit a positive post-pilot coefficient of 0.2270 ($p=0.055$), which may reflect a partial rebound in trading activity for Group 3 stocks following removal of the trade-at prohibition.

In sharp contrast, the negative impact on turnover is heavily concentrated among constrained stocks. The triple interaction coefficient for Group 1 is -0.1691 and statistically significant at the 5 percent level, indicating a meaningful additional reduction in turnover for stocks with pre-treatment spreads below \$0.05 relative to unconstrained stocks within the same group. Crucially, and unlike models that strictly drop entire firms based on post-treatment price fluctuations, our Intention-To-Treat (ITT) framework preserves the necessary sample variance to reveal significant effects across the other groups as well. For Groups 2 and 3, the triple interaction coefficients are -0.1607 and -0.1867 , respectively. Both are economically large and statistically significant at the 10 percent level ($p=0.074$ and $p=0.054$), confirming that the volume reduction among constrained stocks is consistent across all three pilot groups. Turning to the post-pilot period, the results for constrained stocks are more nuanced. For Group 1, the post-pilot triple interaction is -0.1110 and not statistically significant ($p=0.304$), suggesting that the volume reduction was fully reversed for Group 1 constrained stocks once the pilot was unwound. For Groups 2 and 3, however, the post-pilot triple interaction coefficients of -0.2017 ($p=0.074$) and -0.2990 ($p=0.014$) remain statistically significant, indicating that the volume suppression among constrained stocks persisted even after the tick size was restored to \$0.01. These post-pilot effects are larger in absolute magnitude than the active treatment-period coefficients, implying that the disruption to trading patterns for constrained stocks in Groups 2 and 3 was not immediately reversed upon the pilot's conclusion and may reflect lasting changes in market participant behavior or liquidity provision in those stocks.

These findings provide compelling support for Hypothesis 2. The statistically significant effect for constrained stocks across the pilot groups confirms that the volume contraction was concentrated almost entirely among firms for which the \$0.05 tick represented a binding mechanical increase in the spread floor. For such stocks, the regulation mechanically widened spreads relative to their prior equilibrium levels, eliminated marginal trades, and severely reduced observed turnover. The persistence of the volume decline into the post-pilot period for Groups 2 and 3 is a particularly notable finding, as it suggests that the disruption was not purely mechanical in nature: even after the \$0.05 floor was removed, constrained stocks in the stricter pilot groups had not fully recovered their pre-pilot trading activity. The control variables are also consistent with theoretical expectations. The coefficient on Inverse Price is -1.060 and highly significant ($p=0.000$), reflecting the mechanical relationship between price level and effective tick size. The coefficient on Lagged Absolute Return is 7.798 and highly significant ($p=0.000$), confirming that recent price volatility is a strong predictor of trading activity.

The heterogeneous specification therefore proves that the average treatment effects reported in Table 2 (Panel A) are not uniformly distributed across stocks. The decline in trading activity is driven primarily by firms whose pre-treatment spreads were sufficiently narrow for the \$0.05 floor to bind. This pattern is perfectly consistent with the microstructure framework outlined earlier and reinforces the causal interpretation of our estimates.

4.3 Economic Interpretation and Implications

Taken together, the results clearly indicate that the SEC Tick Size Pilot Program reduced trading activity among small capitalization stocks, particularly for firms previously characterized by narrow spreads. The estimated average treatment coefficients, reaching up to -0.0954 , and the specific heterogeneous triple interaction coefficients ranging from -0.1607 to -0.1867 for constrained stocks across the treated groups, are economically large and meaningful.

The evidence is consistent with the hypothesis that artificially widening the minimum tick size acts as a severe friction that increases effective transaction costs and aggressively discourages trade. Although the pilot was partly motivated by the expectation that larger ticks could strengthen market maker incentives and improve liquidity provision, our results decisively suggest that any such theoretical benefits were entirely dwarfed by the contraction in trading activity. Importantly, these findings remain robust when trading activity is alternatively measured using the natural logarithm of trading volume, as evidenced in Table 4 in the Appendix, confirming that the observed contraction is not driven by the scaling of turnover but reflects a genuine decline in trading intensity.

From a policy perspective, the findings imply that increasing the tick size inherently harms trading volume in small-cap equities, especially when spreads were previously compressed below the new regulatory threshold. The results highlight the critical importance of accounting for heterogeneity in pre-existing market conditions when evaluating market microstructure reforms.

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Appendix

Table 3: Equality Test of Treatment Effects Across Groups

Hypothesis Tested	F-statistic	df (1, 1699)	p-value
$G3 \times \text{Treatment} = G1 \times \text{Treatment}$	0.03	(1, 1699)	0.8707
$G3 \times \text{Treatment} = G2 \times \text{Treatment}$	0.41	(1, 1699)	0.5196

The table reports F-tests for equality of treatment-period coefficients across pilot groups. Standard errors are clustered at the firm (permno) level. The null hypothesis in each case is that the treatment effect for Group 3 equals the treatment effect for the comparison group.

Table 4: Regression Results for H1 and H2 (Dependent Variable: $\ln(\text{Volume})$)

Panel A: Regression Results for H1 (Dependent Variable: $\ln(\text{Volume})$)

Variable	Coefficient	Std. Err.	t	P> t	[95% Conf. Interval]
$G1 \times \text{Treatment Period}$	-0.1080746	0.0367108	-2.94	0.003	-0.1800777 -0.0360714
$G2 \times \text{Treatment Period}$	-0.0679551	0.041314	-1.64	0.100	-0.1489868 0.0130765
$G3 \times \text{Treatment Period}$	-0.0930485	0.0400033	-2.33	0.020	-0.1715093 -0.0145876
$G1 \times \text{Post-Pilot}$	-0.0798839	0.0512311	-1.56	0.119	-0.1803666 0.0205989
$G2 \times \text{Post-Pilot}$	-0.111723	0.0559407	-2.00	0.046	-0.221443 -0.002003
$G3 \times \text{Post-Pilot}$	-0.0045916	0.0525001	-0.09	0.930	-0.1075631 0.09838
Inverse Price	-0.8004007	0.1806414	-4.43	0.000	-1.154704 -0.4460977
Lagged Absolute Return	7.880534	0.1302511	60.50	0.000	7.625065 8.136004
Constant	11.24804	0.0197169	570.48	0.000	11.20937 11.28671

Panel B: Regression Results for H2 (Dependent Variable: $\ln(\text{Volume})$)

Variable	Coefficient	Robust Std. Err.	t	P> t	[95% Conf. Interval]
$G1 \times \text{Treatment Period}$	0.0419669	0.0834859	0.50	0.615	-0.121779 0.2057129
$G2 \times \text{Treatment Period}$	0.0823146	0.1001682	0.82	0.411	-0.1141513 0.2787806
$G3 \times \text{Treatment Period}$	0.0677043	0.0949482	0.71	0.476	-0.1185233 0.253932
$G1 \times \text{Post-Pilot}$	0.0773733	0.1159349	0.67	0.505	-0.1500169 0.3047635
$G2 \times \text{Post-Pilot}$	0.1010166	0.1280697	0.79	0.430	-0.1501744 0.3522075
$G3 \times \text{Post-Pilot}$	0.2614307	0.130497	2.00	0.045	0.0054789 0.5173824
$G1 \times \text{Treatment} \times \text{Small Spread}$	-0.2020722	0.0863302	-2.34	0.019	-0.3713969 -0.0327476
$G2 \times \text{Treatment} \times \text{Small Spread}$	-0.1994962	0.1043368	-1.91	0.056	-0.4041383 0.0051459
$G3 \times \text{Treatment} \times \text{Small Spread}$	-0.2141391	0.0987925	-2.17	0.030	-0.4079068 -0.0203713
$G1 \times \text{Post-Pilot} \times \text{Small Spread}$	-0.211742	0.1211839	-1.75	0.081	-0.4494274 0.0259435
$G2 \times \text{Post-Pilot} \times \text{Small Spread}$	-0.2830405	0.1352627	-2.09	0.037	-0.5483396 -0.0177415
$G3 \times \text{Post-Pilot} \times \text{Small Spread}$	-0.3553413	0.1343266	-2.65	0.008	-0.6188043 -0.0918783
Inverse Price	-0.7996322	0.1791522	-4.46	0.000	-1.151014 -0.4482499
Lagged Absolute Return	7.864341	0.1300125	60.49	0.000	7.609339 8.119342
Constant	11.24791	0.0195554	575.18	0.000	11.20956 11.28627

The dependent variable is $\ln(\text{Volume})$ in both Panels A and B. $G1$, $G2$, and $G3$ indicate assignment to the SEC Tick Size Pilot treatment groups. Treatment equals one during the pilot implementation period. SmallSpread equals one if the firm's pre-treatment average quoted spread is below \$0.05. All regressions include firm and date fixed effects. Standard errors are clustered at the firm level and reported in parentheses. The table reports t-statistics, p-values, and 95% confidence intervals.